Hough Transform:

Example 1

Consider a 6x6 binary image matrix with pixels set to 1 at points (2, 1) (4, 3) and (6, 5) other pixels set to 0.

$$\begin{matrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{matrix}$$

$P_1 = (2, 1)$

$P_2 = (4, 3)$

$P_3 = (6, 5)$

Compute both cartesian equation & HT.

Step 1: Straigh line (cartesian).

i) calculate slope m.

Take slope b/w P_1 & P_2 is

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3-1}{4-2} = \frac{2}{2} = 1$$

$\therefore m = 1$

ii) Use point (2,1) ~~(4,3) (6,5)~~ find intercept

Using $P_1 = (\overset{x}{2}, \overset{y}{1})$

$y = mx + c$

$1 = 1 \cdot 2 + c$

$c = -1$

$P_2 = (4,3)$

$y = mx + c$

$3 = 1 \cdot 4 + c$

$3 = 4 + c$

$c = -3$

$P_3 = (6,5)$

$y = mx + c$

$5 = 1 \cdot 6 + c$

$5 = 6 + c$

$c = -1$

iii) In Cartesian equation apply m & c

$y = mx + c$

$y = 1 \cdot x + (-1)$

$\boxed{y = x - 1}$

Now,

$P_1 (\overset{x}{2}, \overset{y}{1})$

$y = x - 1$

$1 = 2 - 1$

$1 = 1$

$P_2 (4,3)$

$y = x - 1$

$3 = 4 - 1$

$3 = 3$

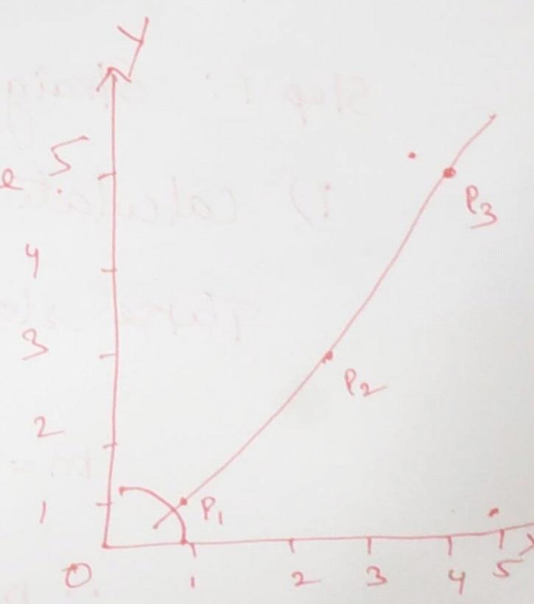
$P_3 (6,5)$

$y = x - 1$

$5 = 6 - 1$

$5 = 5.$

All lie on the Same line



Step 2: polar or parametric form.

We need to find θ & perpendicular (ρ) or (r).

$$r = x\cos\theta + y\sin\theta$$

Start from 0° to 180°.

We will take 45°.

$$\theta = 45^\circ$$

$$\cos\theta(45^\circ) = 0.707$$

$$\sin(45^\circ) = 0.707$$

Compute ρ or r for P_1 $(\overset{x}{2}, \overset{y}{1})$

$$r = 2\cos(45^\circ) + y\sin(45^\circ)$$

$$= 2(0.707) + 1(0.707)$$

$$\rho_1 \text{ or } r_1 = 2.121$$

P_2 $(4, 3)$

$$r_2 = 4(0.707) + 3(0.707)$$

$$= 4.949$$

P_3 $(6, 5)$

$$r_3 = 6(0.707) + 5(0.707)$$

$$= 7.778.$$

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$\theta = 135^\circ$

$\cos(135^\circ) = -0.707$

$\sin(135^\circ) = 0.707$

Calculate ρ or r.

For P_1 $(2,1)$

ρ_1 or $r_1 = 2(-0.707) + 1(0.707)$

$= -1.414 + 0.707$

$= -0.707$

P_2 $(4,3)$

$r_2 = \cancel{-0.707}$

$4(-0.707) + 3(0.707)$

$= -2.828 + 2.121$

$= -0.707$

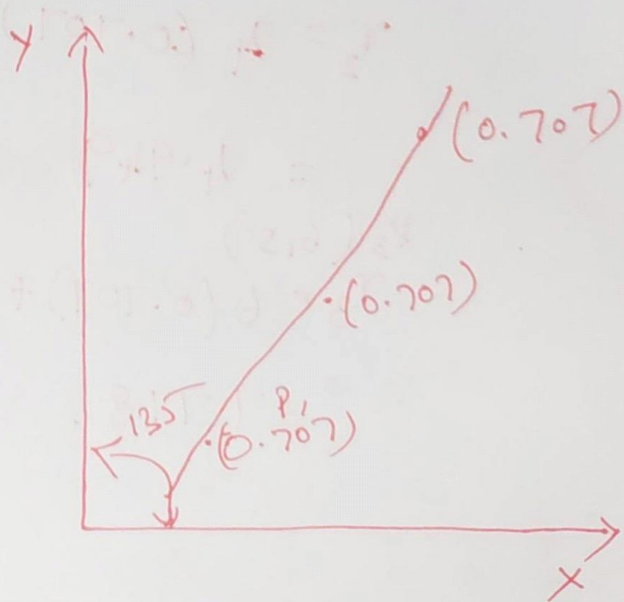
P_3 $(6,5)$

$r_3 = 6(-0.707) + 5(0.707)$

$= -4.242 + 3.535$

$= -0.707$

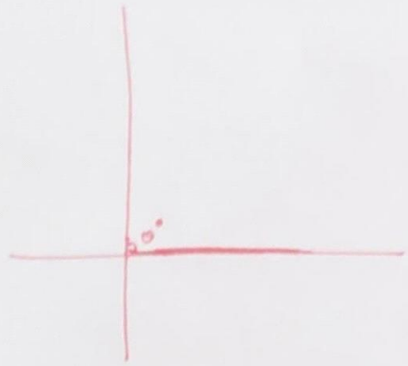
$\theta = 135^\circ$



Example 2 :

6x6 Binary Image as input.

0	0	0	0	0	0
0	0	0	0	0	0
0	0	0	0	0	0
0	1	0	0	0	1
0	0	0	0	0	0
0	0	0	0	0	0



2 Edge point : (4, 2) (4, 6)

Compute Slope & Intercept (Cartesian Equation)

Step 1 : calculate slope M.

$P_1 (\overset{x_1}{4}, \overset{y_1}{2})$ $P_2 (\overset{x_2}{4}, \overset{y_2}{6})$

$$m = \frac{y_2 - y_1}{x_2 - x_1} \Rightarrow \frac{6-2}{4-4} \Rightarrow 4/0 \Rightarrow 0$$

↓

Need to find out polar form

$$r = x\cos\theta + y\sin\theta$$

compute (ρ, θ) for each point :

(4, 2)

$\theta = 90^\circ$

$$\rho = 4\cos(90^\circ) + 2\sin(90^\circ)$$
$$\Rightarrow 0 + 2 \Rightarrow 2$$

$\theta = 0^\circ$

$$\rho = 4\cos(0^\circ) + 2(\sin 0^\circ)$$
$$= 4 + 0 \Rightarrow 4$$

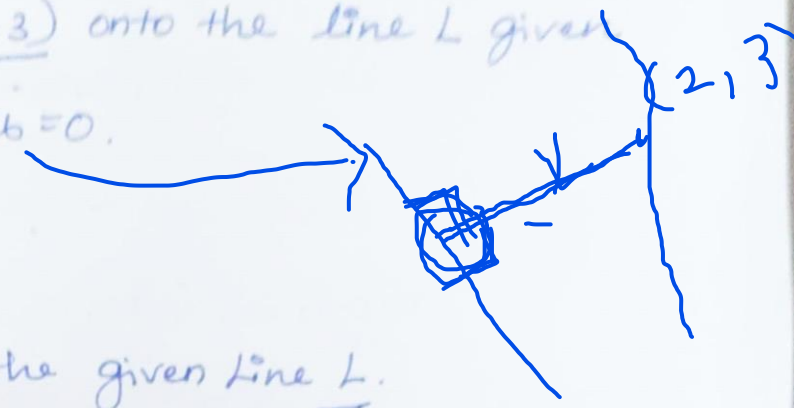
(4, 6)

$$\rho = 4(\cos 90^\circ) + 6(\sin 90^\circ)$$
$$= 6$$

$$\rho = 4\cos(0^\circ) + 6\sin 0^\circ$$
$$= 4 + 0$$
$$\Rightarrow 4.$$

Foot of Normal Method

1. Find the coordinates of the foot of the perpendicular (F) from the external point P(2,3) onto the line L given by the equation $3x - 4y - 16 = 0$.



Method 1: (cartesian)

Step 1: Find the slope of the given Line L.

The equation of L is $3x - 4y - 16 = 0$.

Convert to slope-intercept form $(y = mx + c)$:

$$4y = 3x - 16$$

$$y = \overset{m}{\frac{3}{4}}x \overset{c}{-4}$$

The slope of line L is $m_L = \frac{3}{4}$

Step 2: Find slope of the perpendicular line PF.

m_{PF} is the negative reciprocal of m_L.

$$m_{PF} = -\frac{1}{m_L} = -\frac{1}{3/4} = -4/3.$$

Step 3: Find the equation of the perpendicular line PF.

Use point-slope $y - y_1 = m(x - x_1)$ with $P(2,3)$ & slope $m_{PF} = -4/3$.

$$y - y_1 = m_{PF}(x - x_1)$$

$$y - 3 = -4/3(x - 2)$$

$$3y - 9 = -4x + 8$$

$$4x + 3y - 17 = 0 \text{ (equation of PF)}$$

Step 4: Find the intersection point F.

solve equ of 2 lines

1. $3x - 4y = 16 \rightarrow (1)$
2. $4x + 3y = 17. \rightarrow (2)$

Now, multiply (1) by 3 and (2) by 4:

$(1) \Rightarrow 3 \times (3x - 4y = 16) = 9x - 12y = 48$

$(2) \Rightarrow 4 \times (4x + 3y = 17) = 16x + 12y = 68$

Now add (1) & (2)

$$\begin{array}{r} 9x - 12y = 48 \\ 16x + 12y = 68 \\ \hline 25x + 0y = 116 \end{array}$$

$$\therefore x = \frac{116}{25}.$$

Substitute x value in equ 2

$$4\left(\frac{116}{25}\right) + 3y = 17$$

$$\frac{464}{25} + 3y = 17$$

isolate y

$$3y = 17 - \frac{464}{25}$$

$$= \frac{17 \times 25 - 464}{25}$$

$$= \frac{425 - 464}{25} = -\frac{39}{25}$$

$$y = -\frac{39}{25} \times \frac{1}{3} = -\frac{13}{25}$$

$\therefore$ F (foot of perpendicular) is

$$F\left(\frac{116}{25}, -\frac{13}{25}\right)$$

Method 2: Direct Formula method.

$P(x_1, y_1)$ & line $Ax+By+C=0$ the coordinates of F (Foot of perpendicular) $F(x, y)$ is given by:

$$\frac{x-x_1}{A} = \frac{y-y_1}{B} = -\frac{Ax_1+By_1+C}{A^2+B^2}$$

Step 1: Identify Values

$P(x_1, y_1) \Rightarrow (2,3)$

$Ax+By+C=0 \Rightarrow 3x-4y-16=0$

Coefficients: $A=3$, $B=-4$, $C=-16$.

Step 2: calculate constant term k

to calculate third part of the formula, we can call k:

$$k = -\frac{Ax_1+By_1+C}{A^2+B^2}$$

$$= -\frac{3(2)+(-4)(3)+(-16)}{3^2+(-4)^2}$$

$$k = -\frac{6-12-16}{9+16}, \quad k = -\frac{-22}{25}$$

$$\therefore k = \frac{22}{25}$$

Step 3: Find x and y of F

Set first two parts of formula equal to k.

For x:

$$\frac{x - x_1}{A} = k$$

$$\frac{x-2}{3} = \frac{22}{25}$$

$$x - 2 = 3 \times \frac{22}{25} = \frac{66}{25}$$

$$x = 2 + \frac{66}{25} = \frac{50}{25} + \frac{66}{25} = \overset{x}{\frac{116}{25}}$$

For y:

$$\frac{y - y_1}{B} = k$$

$$\frac{y-3}{-4} = \frac{22}{25}$$

$$y - 3 = -4 \times \frac{22}{25} = -\frac{88}{25}$$

$$y = 3 - \frac{88}{25} = \frac{75}{25} - \frac{88}{25} = \overset{y}{-\frac{13}{25}}$$

$\therefore$ Foot of perpendicular $F\left(\overset{x}{\frac{116}{25}}, \overset{y}{-\frac{13}{25}}\right)$

Line Localization

A binary image provides the following three edge points:

$P_1 = (1, 4)$

$P_2 = (2, 3)$

$P_3 = (3, 2)$

find

1. Define a discretized parameter space (ρ, θ)
2. voting (calculating ρ and accumulating)
3. Accumulator Array (Final Tally)
4. Line Identification (Peak Finding)

Step 1: Define parameter space (ρ, θ)

We need to discretize the hough space (ρ, θ)

→ Lets test a few common angle(θ) like 45°, 90°, 135° relate to vertical, diagonal, horizontal.

→ Distance (ρ) : from 1 to 6 (based on coordinates).

- We create an Accumulator Array (table) to all zeros.

$\rho \backslash \theta$	45°	90°	135°
1	0	0	0
2	0	0	0
3	0	0	0
4	0	0	0
5	0	0	0
6	0	0	0

Step 2: Voting (calculating ρ and accumulating)

→ line equation $\rho = x\cos\theta + y\sin\theta$)

We will use $\cos 45^\circ = 0.707$, $\sin 45^\circ = 0.707$

$\cos 90^\circ = 0$, $\sin 90^\circ = 1$

$\cos 135^\circ = -0.707$, $\sin 135^\circ = 0.707$

A. point $P_1 = (1,4)$

θ	$\rho = \overset{1}{x}\cos\theta + \overset{4}{y}\sin\theta$	ρ (rounded)	Votes Added
45°	$1(0.707) + 4(0.707) = 3.535$	4	+1 at ($\rho=4, \theta=45^\circ$)
90°	$1(0) + 4(1) = 4$	4	+1 at ($\rho=4, \theta=90^\circ$)
135°	$1(-0.707) + 4(0.707) = 2.121$	2	+1 at ($\rho=4, \theta=135^\circ$)

B. point $P_2(2,3)$

θ	$\rho = \overset{2}{x}\cos\theta + \overset{3}{y}\sin\theta$	ρ (rounded)	Votes Added
45°	$2(0.707) + 3(0.707) = 3.535$	4	+1 at $\rho=4, \theta=45^\circ$
90°	$2(0) + 3(1) = 3$	3	+1 at $\rho=3, \theta=90^\circ$
135°	$2(-0.707) + 3(0.707) = 0.707$	1	+1 at $\rho=1, \theta=135^\circ$

C. point $P_3(3,2)$

θ	$\rho = \overset{3}{x}\cos\theta + \overset{2}{y}\sin\theta$	ρ (rounded)	Votes Added
45°	$3(0.707) + 2(0.707) = 3.535$	4	+1 at ($\rho=4, \theta=45^\circ$)
90°	$3(0) + 2(1) = 2$	2	+1 at ($\rho=2, \theta=90^\circ$)
35°	$3(-0.707) + 2(0.707) = -0.707$	-1 (outside our range) so no vote.	No vote

Step 3: Accumulator Array (Final Tally)

After all 3 points have voted,

$\rho \backslash \theta$	45°	90°	135°
1	0	0	1
2	0	1	1
3	0	1	0
4	3	1	0
5	0	0	0
6	0	0	0

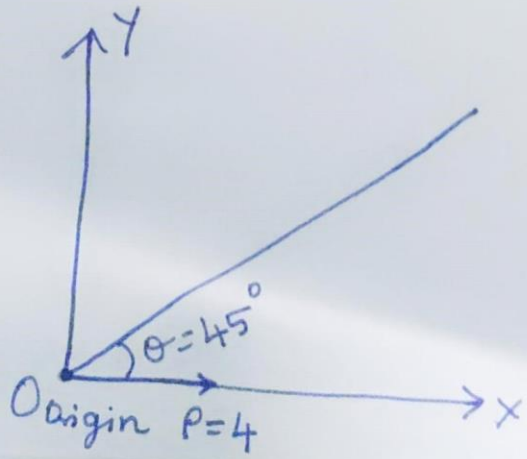
Step 4: Line Identification (Peak finding)

Now, Look for the cell with highest vote count.

The peak is at $(\rho = 4, \theta = 45^\circ)$ wic is a count of 3.

$\therefore$ Detected Line : $(\rho = 4, \theta = 45^\circ)$

This means the detected line is located 4 units from origin & has perpendicular angle of 45°.



Line fitting

Solution:

Study Hours(x)	Test Score(y)
1	3
2	4
4	7
5	6

Sol)

The line eqn is $y = ax+b$, and the no. of datapoints is $n=4$.

Step 1: Calculate the necessary sums

create table for Σx, Σy, Σx^2 & Σxy.

	x	y	x^2	xy
	1	3	1	3
	2	4	4	8
	4	7	16	28
	5	6	25	30
Σ	12	20	46	69

$\therefore \Sigma x = 12$

$\Sigma y = 20$

$\Sigma x^2 = 46$

$\Sigma xy = 69$

$n = 4$.

Step 2: calculate slope (a)

$$a = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{n(\Sigma x^2) - (\Sigma x)^2}$$

$$a = \frac{4(69) - (12)(20)}{4(46) - (12)^2}$$

$$a = \frac{276 - 240}{184 - 144}, \quad a = \frac{36}{40} = 0.9$$

The slope is $a = 0.9$

Step 3: calculate the Y-intercept (b)

find mean

$$\bar{x} = \frac{\Sigma x}{n} = \frac{12}{4} = 3$$

$$\bar{y} = \frac{\Sigma y}{n} = \frac{20}{4} = 5$$

use formula for b:

$$b = \bar{y} - a\bar{x}$$

$$b = 5 - (0.9)(3)$$

$$b = 5 - 2.7 = 2.3$$

The y-intercept is $b = 2.3$

Step 4: Write the final eqn.

Substitute a & b value into linear eqn.

$$Y = ax + b.$$

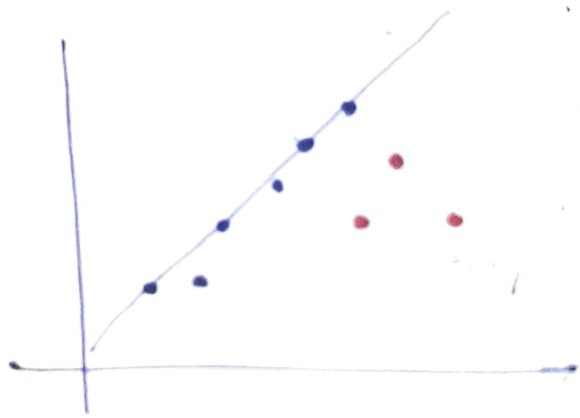
$$\therefore y = 0.9x + 2.3$$

RANSAC - Random Sample Consensus

It is an iterative Algorithm used in data fitting to estimate parameters of a mathematical model from data containing outliers

It works by randomly selecting a small subset of the data, fitting a model to it, and then counting how many data points fit that model (inliers).

This process is repeated multiple times, and the model that results in the highest number of inliers is selected as the best fit, effectively robustly handling outliers.



- Inliers (correct edge points)
- outliers (Noise, Incorrect edge points)

Inliers : Data points that conform to the underlying model.

outliers : Data points that do not fit the model, often due to errors or noise in the data

example :

i) object detection

ii) Robotics

Example:

Edge points (1,2), (2,3), (3,4), (4,5), (5,20), (6,30)

Step 1: Random Sampling:

1) Randomly choose 2 points (since 2 points define a line)

eg choose (1,2) and (3,4)

Step 2: Model fitting:

Fit a line $\boxed{y = mx + c}$

1, 2
x_1 y_1

3, 4
x_2 y_2

Slope $m = \frac{y_2 - y_1}{x_2 - x_1} \Rightarrow \frac{4-2}{3-1} \Rightarrow 2/2 \Rightarrow 1$

$\boxed{m = 1}$

(1, 2)
x y

Intercept

$y = mx + c$

$c = y - mx$

$= 2 - 1(1)$

$= 2 - 1$

$\boxed{c = 1}$

Equation $\boxed{y = x + 1}$

Step 3: compute outliers:

Set Threshold $T = 1$

compute distance of all points to the line.

m, c

Distance from point (x_i, y_i) to line $y = mx + c$

$$\boxed{d_i = \frac{|y_i - (mx_i + c)|}{\sqrt{m^2 + 1}}}$$

(1, 2)

$\frac{2 - (1+1)}{\sqrt{2}}$

0

point (1, 2)

$y = 2$

predicted y on line $y = mx + c$
$= 1 \cdot 1 + 1$
$= 1 + 1$
$\Rightarrow 2$

$|y_i - (mx_i + c)| = |2 - 2|$
$= 0$

$\sqrt{m^2+1}$
$\Rightarrow \sqrt{1^2+1}$
$= \sqrt{2}$
$= 1.414$

distance $= \frac{0}{\sqrt{2}} \Rightarrow 0/1.414$
$\Rightarrow 0$

Threshold $T = 1$ since $0 \leq 1 \Rightarrow$ $\boxed{\text{Inlier}}$ ✓

Repeat for all the points.

2, 3

$y = 3$

predicted y on line $y = mx + c$
$= 1 \cdot 2 + 1$
$= 2 + 1$
$\Rightarrow 3$

$|y_i - (mx_i + c)| \Rightarrow |3 - (3)|$
$= 3 - 3$
$\Rightarrow 0$.

$\sqrt{m^2+1} \Rightarrow \sqrt{1^2+1} = \sqrt{2}$
$= 1.414$

distance $= 0/\sqrt{2} = 0/1.414$
$\Rightarrow 0$.

Threshold $T = 1$ since $0 \leq 1 \Rightarrow$ $\boxed{\text{Inlier}}$ ✓

(3,4)

$y = 4$

Predicted y on line $y = mx + c$

$= 1.3 + 1$

$= 3 + 1$

$y \Rightarrow 4$

$|y_i - (mx_i + c)| = |4 - 4| = 0$

$\sqrt{m^2+1} = \sqrt{1^2+1} = \sqrt{2} = 1.414$

$d = 0/\sqrt{2} = 0/1.414 \Rightarrow 0$

Threshold $T = 1$ since $0 \leq 1 \Rightarrow$ $\boxed{\text{Inlier}}$

(4,5)

$y = 5$

predicted y on line $y = mx + c$

$= 1.4 + 1$

$y = 5$

$|y_i - (mx_i + c)| \Rightarrow |5 - (5)| \Rightarrow 0$

$\sqrt{m^2+1} = \sqrt{1^2+1} = \sqrt{2} = 1.414$

$d = 0/1.414 \Rightarrow 0$

Threshold $T = 0$ since $0 \leq 1 \Rightarrow$ $\boxed{\text{Inlier}}$

5,20

$y = 20$

predicted y on line $y = mx + c$

$= 1.5 + 1$

$y = 6$

$|y_i - (mx_i + c)| = |20 - (6)| \Rightarrow 14$

$\Rightarrow \sqrt{1^2+1} = \sqrt{2} = 1.414$

$d = 14/1.414 \Rightarrow 9.899$

Threshold $T = 1$ since $0 \leq 1 \Rightarrow$ $\boxed{\text{outlier}}$ because 9.899

6, 30

$y = 30$

Predicted y on line $y = mx + c$

$= 1 \cdot 6 + 1$

$\Rightarrow 7$

$|y_i - (mx_i + c)| = |30 - 7| \Rightarrow 23$

$\sqrt{1^2 + 1} = \sqrt{2} = 1.414$

$d = 23 / 1.414 \rightarrow 16.265$

Threshold $T = 1$ since $0 \leq 1 \Rightarrow$ $\boxed{\text{outliers}}$ because of 16.265

Step : 4 : Build consensus set :

All points with distance $T \leq 1$ are Inliers

$\boxed{(1,2), (2,3), (3,4), (4,5)} \rightarrow$ Inliers

$\boxed{(5,20), (6,30)} \rightarrow$ outliers.

Step 5 : Repeat Iterations :

Again pick 2 points (2,3) (3,4)

Repeat step (1) to step (4) for finding Inliers and outliers.

Step 6: Final Refit using inliers:

Least Square fit: Inliers are

1, 2

2, 3

3, 4

4, 5

$$m = \frac{N\sum x_i y_i - \sum x_i \sum y_i}{N\sum x_i^2 - (\sum x_i)^2}$$

$$c = \frac{\sum y_i - m\sum x_i}{N}$$

x	y	$\sum x_i^2$	$\sum x_i y_i$
1	2	1^2	$1 \cdot 2$
2	3	2^2	$2 \cdot 3$
3	4	3^2	$3 \cdot 4$
4	5	4^2	$4 \cdot 5$
$\sum x = 10$	$\sum y = 14$	$\sum x_i^2 = 30$	40

$$m = \frac{4 * 40 - 10 * 14}{4 * 30 - (10)^2} \Rightarrow \frac{160 - 140}{120 - 100} \Rightarrow \frac{20}{20} = 1$$

$$c = \frac{14 - 1 * 10}{4} \Rightarrow \frac{14 - 10}{4} = \frac{4}{4} \Rightarrow 1$$

$m = 1$

$c = 1$

$y = mx + c$

$y = 1 \cdot x + 1$

$$\boxed{y = x + 1}$$

This line fits all inliers perfectly.